

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 3: Observe (Part 2): Lipids and Vitamins — Structure, Function, and Food Tests

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	B	C	B	A	B	C

USE AFTER: OBSERVE PHASE — Activity 1: Translucent Spot Test

Q1. A learner rubs mashed groundnuts on a piece of brown paper bag and waits 5 minutes for it to dry. Which observation shows that the groundnuts contain lipids?

- A. A translucent (greasy) spot stays on the paper even after it is dry**
- B. A wet spot appears but disappears completely once the paper dries
- C. The paper stays completely dry and unmarked
- D. The paper turns a purple colour where the food was rubbed

Correct answer: A. Lipids leave a permanent translucent spot that does not disappear on drying. Water-based foods leave a spot that dries and disappears.

USE AFTER: OBSERVE PHASE — Activity 2: Ethanol Emulsion Test

Q2. In the ethanol emulsion test, a learner dissolves a food sample in ethanol, then pours the clear ethanol layer into a test tube of distilled water. What result shows that lipid is present?

- A. The mixture stays clear and colourless
- B. A cloudy white emulsion forms**
- C. The mixture bubbles and gives off heat
- D. The mixture turns purple

Correct answer: B. Lipids dissolve in ethanol but not in water, so when the ethanol is added to water the lipid comes out as tiny droplets, forming a cloudy white emulsion. A clear solution means no lipid.

USE AFTER: OBSERVE PHASE — Activity 1 and Activity 2: results table for ugali (A)

Q3. A group tests ugali paste (maize starch) with both the translucent spot test and the ethanol emulsion test. Which set of results would you expect?

- A. A permanent translucent spot and a cloudy white emulsion
- B. No permanent translucent spot but a cloudy white emulsion
- C. No permanent translucent spot and a clear solution in water**
- D. A permanent translucent spot but a clear solution in water

Correct answer: C. Ugali is mostly starch and has very little lipid, so it should give a negative result in both tests. Two tests that agree make the evidence stronger.

USE AFTER: OBSERVE PHASE — Activity 3: Guided Reading on Vitamins and the Vitamin Summary Strip

Q4. Which row correctly matches a vitamin with its type and the deficiency disease it causes?

- A. Vitamin C: fat-soluble; deficiency causes scurvy
- B. Vitamin A: fat-soluble; deficiency causes night blindness**
- C. B-complex vitamins: fat-soluble; deficiency causes beriberi
- D. Vitamin D: water-soluble; deficiency causes rickets

Correct answer: B. Vitamins A, D, E and K are fat-soluble, and lack of vitamin A causes night blindness. Vitamin C and the B-complex vitamins are water-soluble, and vitamin D is fat-soluble.

USE AFTER: EXPLAIN PHASE — Whole-class discussion: essential fatty acids and fat-soluble vitamin absorption, and the Temperature Check on night blindness

Q5. A child eats a meal rich in vitamin A but has almost no fat in their diet. Why might the child still develop night blindness?

- A. Vitamin A is fat-soluble and needs dietary fat to be absorbed**
- B. Vitamin A is a protein, so it cannot be used without amino acids
- C. Fat destroys vitamin A in the stomach
- D. Vitamin A is water-soluble and is washed out of the body too quickly

Correct answer: A. Vitamins A, D, E and K are fat-soluble and are absorbed only when fat is present in the diet. Without fat, the body cannot fully absorb vitamin A even from a vitamin-rich meal.

USE AFTER: MODEL BUILDING PHASE — Revising the Cumulative Chemical Model of Food: 'Lipids' section showing structure

Q6. At the chemical level, what is a lipid (fat) made of?

- A. Vitamins joined to a mineral salt
- B. A glycerol backbone joined to fatty acid chains**
- C. Long chains of amino acids joined together
- D. Many glucose units joined in a long chain

Correct answer: B. A lipid such as a fat is made of one glycerol molecule joined to three fatty acids. Amino acid chains are proteins and glucose chains are starch.

USE AFTER: END OF LESSON — End of lesson: after the Cumulative Chemical Model revision and peer feedback

Q7. Which statement correctly describes the roles of lipids in living organisms?

- A. Lipids are harmful chemicals that the body does not need
- B. Lipids only provide vitamins and have no other role
- C. Lipids store energy and are part of cell membranes**
- D. Lipids are the main chemicals used to build muscles and repair tissues

Correct answer: C. Lipids store energy, form part of cell membranes and help absorb fat-soluble vitamins. Building and repairing muscles is mainly the role of proteins, and calling fat 'unhealthy' is a common misconception.